

$$\begin{array}{ccc}
 \mathrm{SL}_2(\mathbb{R}) & \xrightarrow{\quad F \quad} & \Omega M_g \\
 \downarrow & & \downarrow \pi \\
 \mathrm{SL}_2(\mathbb{R})/\mathrm{SO}_2(\mathbb{R}) \cong \mathbb{H} & \dashrightarrow_f & M_g
 \end{array}$$